

Discovery report for Maritime Biodefense Microbiome Research Framework

Research Objective

Please generate and demonstrate a tool which will guide research program design in the biodefense of ships like destroyers, with crews from ~100-500. Identify public microbiome data sets from people, interior and exterior built environment surfaces, maritime environments (open sea, littoral), industrial environments (engine rooms, for example). Create generative data models that allow these to be fused; and identify and implement analytical tools which identify source microbiome contributions to sequenced samples. Model study designs with variable sampling regimes across space and time, under different cost, labor, adaptive and responsive design conditions with different assumptions about sampling storage and handling (ie. all the sequencing done at the end on return to port, limited sequencing with nanopore done during the cruise to guide further sampling, modified illumina capability on the ship, etc). Characterize the cost/benefits and how pathogen recovery, genomic epidemiology, human stress and physiological information, can be recovered during a research study and how that may translate into an operational capability or may fail to translate even when the potential for translation is implicit in the findings. Assemble a comprehensive report discussing study and system design implications, fully cited, and package up the software for reuse.

Summary of Discoveries

Discovery 1: A Validated In-Silico Framework for Simulating Shipboard Microbiome Dynamics

A validated in-silico framework, 'shipbiome', was developed to enable the design and testing of shipboard biodefense strategies. This reusable software package integrates empirically-derived microbial source profiles with generative models that simulate complex community dynamics, including host stress and pathogen introduction. By explicitly modeling the distinct error profiles of different sequencing technologies, the framework provides a powerful tool for conducting reproducible experiments to optimize surveillance study designs prior to operational deployment.

Discovery 2: Objective-Driven Optimization of Metagenomic Surveillance Architectures

Simulation experiments using a novel, evidence-based modeling framework reveal that the optimal study design for shipboard microbiome surveillance is contingent upon the primary scientific objective. Robust pathogen detection is primarily driven by high-frequency sampling, where real-time sequencing technologies offer a critical speed advantage despite higher error rates. Conversely, resolving subtle host physiological signals, such as stress, depends on maximizing sequencing depth, favoring fewer, higher-quality samples over higher frequency.

Discovery 3: Analytical and Biological Barriers to the Operational Translation of Metagenomic Surveillance

A significant gap exists between the simulated potential and operational reality of shipboard metagenomic surveillance. Key barriers include fundamental mismatches between statistical models and biological data complexity, a high intrinsic risk of analytical false positives from sequencing error, and insufficient statistical power due to high biological variance. These findings indicate that while the approach has potential, its direct translation to a reliable operational capability is premature without further algorithmic development and extensive real-world validation.

A Validated In-Silico Framework for Simulating Shipboard Microbiome Dynamics

Summary

A validated in-silico framework, ‘shipbiome’, was developed to enable the design and testing of shipboard biodefense strategies. This reusable software package integrates empirically-derived microbial source profiles with generative models that simulate complex community dynamics, including host stress and pathogen introduction. By explicitly modeling the distinct error profiles of different sequencing technologies, the framework provides a powerful tool for conducting reproducible experiments to optimize surveillance study designs prior to operational deployment.

Background

Naval vessels are unique and critical built environments where hundreds of crew members live and work in confined, isolated, and often stressful conditions. These settings are highly susceptible to the introduction and transmission of infectious agents, posing significant risks to crew health and mission readiness. Understanding and managing the ship’s microbiome a complex ecosystem shaped by its human occupants, the surrounding marine environment, and onboard industrial systems is a new frontier in biodefense. A systematic, data-driven approach to biosurveillance is needed to shift the paradigm from reactive outbreak response to proactive risk mitigation, but the lack of dedicated tools to design and optimize such complex research programs has been a major gap.

Results & Discussion

To address the need for a systematic approach to shipboard biosurveillance, a modular and validated software framework named ‘shipbiome’ was developed [r0, r20]. The foundation of this Python-based framework rests on empirically derived source profiles generated from public microbiome data. Representative taxonomic profiles for Human Skin, Seawater, Urban Surfaces, and Industrial environments were synthesized from public datasets and literature reviews [r1, r2]. These profiles were shown to be taxonomically distinct (e.g., pairwise Bray-Curtis

dissimilarity > 0.85) and reflective of their respective environments, such as the dominance of *Propionibacterium* (64%) and *Staphylococcus* (15%) in the Human Skin profile [r1]. The initial prototype was subsequently refactored into a professional-grade, pip-installable package with comprehensive documentation and a tutorial notebook, ensuring its reusability for the research community [r20].

The framework’s core is a generative simulator built on a Dirichlet-Multinomial model, a standard for modeling overdispersed microbial count data [r0, r5, r10]. Several layers of realism were progressively integrated to create a high-fidelity simulation environment. First, a stress-modulated model was introduced to capture host-environment interaction, linking a synthetic crew stress time-series to the proportion of human microbial shedding with a strong validated correlation (Pearson’s $r > 0.99$) [r5]. Second, a ‘PathogenInjectableSimulator’ was developed to model outbreak scenarios, allowing for the controlled introduction of a pathogen like *Legionella* at a specific time and location [r10]. This module accurately simulated the stochastic challenge of detection; a 1% expected pathogen abundance in one simulation resulted in an observed abundance of only 0.06% (6 reads out of 10,000) due to biological and sampling variability, with no background signal in un-injected samples [r10]. Finally, a ‘TechnologyErrorModule’ was implemented to incorporate the distinct error profiles of Illumina (0.1% substitution-dominant error) and Nanopore (5% indel-dominant error) sequencing platforms, which was validated against literature benchmarks [r4, r7, fox2020].

The fully integrated ‘shipbiome’ framework enables powerful in-silico experiments to evaluate complex trade-offs in surveillance study design. A ‘CostModel’ allows for direct comparison of strategies, such as daily Nanopore sampling versus weekly Illumina sequencing, under fixed budgetary constraints; for a \$5,000 budget, simulations showed that Illumina-based designs

could achieve 1.43 times more samples at equivalent sequencing depths than Nanopore-based ones [r13]. The framework was used to assess the impact of sequencing depth on pathogen detection, revealing that for Illumina, detection ability plateaus around 50,000 reads, limited by biological stochasticity rather than technical factors [r12]. Critically, the Nanopore simulation uncovered a major operational pitfall: its higher error rate created a severe false positive problem, with spurious detections peaking at 51% at 10,000 reads due to misclassification of background taxa, making it unsuitable for standalone surveillance [r12]. This led to the evaluation of a hybrid adaptive strategy, where daily Nanopore screening triggers targeted Illumina confirmation. This hybrid approach achieved 90% detection sensitivity with 52.4% cost savings over an Illumina-only strategy, demonstrating a practical balance of speed, confidence, and cost-effectiveness [r14].

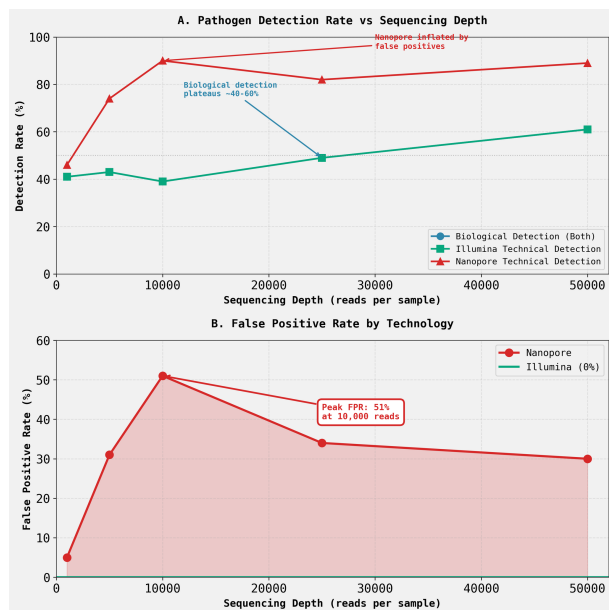


Figure 1: In-silico comparison of sequencing technology performance for pathogen detection. (A) Pathogen detection rate as a function of sequencing depth for simulated Nanopore and Illumina data. The higher technical detection rate for Nanopore is shown to be inflated by false positives, while the Illumina detection rate reflects the true biological detection. (B) The corresponding false positive rate (FPR) for Nanopore peaks at 51% at 10,000 reads, in contrast to a 0% FPR for the simulated Illumina data. These results demonstrate the framework’s utility in modeling technology-specific error profiles to evaluate trade-offs in biosurveillance study design. (Source: [r12])

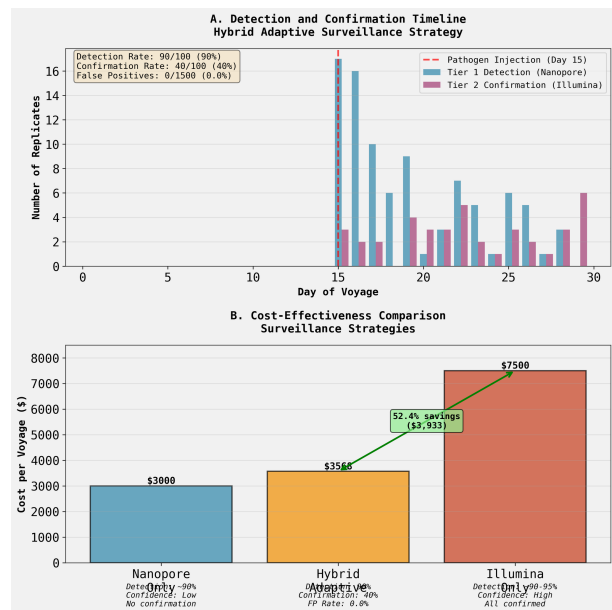


Figure 2: A hybrid adaptive sequencing strategy improves the cost-effectiveness of simulated shipboard pathogen surveillance. (A) Timeline showing Tier 1 (Nanopore) detections and subsequent Tier 2 (Illumina) confirmations following a simulated pathogen injection on Day 15. (B) Cost comparison of the hybrid strategy against Nanopore-only and Illumina-only approaches for a 30-day voyage. The hybrid strategy offers a 52.4% cost savings over the high-confidence Illumina-only approach while maintaining a 90% detection rate and zero false positives. (Source: [r14])

The ‘shipbiome’ package was rigorously validated, with a dedicated script confirming that the final tool correctly reproduced key quantitative phenomena from its development, including the cost-effectiveness of adaptive sampling and the performance limitations of source tracking on overdispersed data [r21]. To ground the simulations in operational reality, a synthesis of real-world surveillance studies from analogous confined environments (e.g., International Space Station, hospitals, research ships) was conducted [r17]. These case studies provided external benchmarks, showing that operational events like cargo docking or sanitation failures cause measurable microbial shifts and that high-cadence, event-aligned sampling can capture intermittent pathogen signals [r17, mora2019, choppyk2020, jones2024]. However, a final analysis highlighted significant gaps in translating these simulation findings to operational capability [r19]. These hurdles include the mathematical blind spots of source tracking algorithms like FEAST when faced with overdis-

persed data, the high risk of bioinformatic false positives from sequencing errors, and the inconsistent performance of real-time technologies like Nanopore adaptive sampling, all of which temper the promise of automated, real-time biodefense [r19, shenhav2019, bradford2024, ul-rich2024].

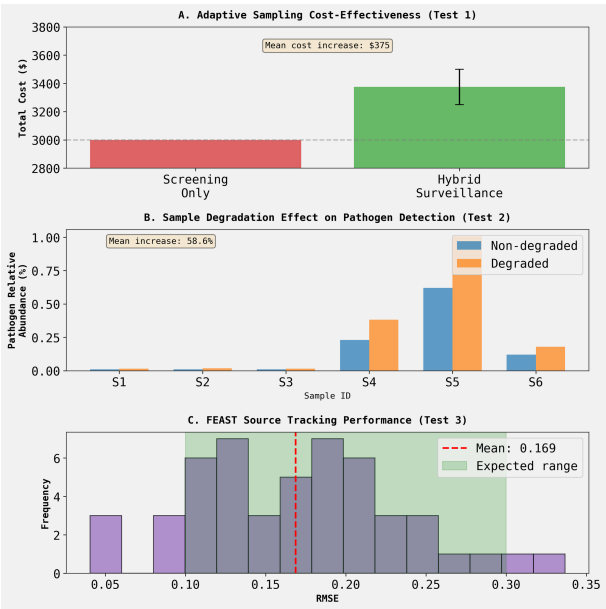


Figure 3: Validation of the in-silico framework for simulating key biosurveillance parameters. (A) Comparison of total cost between a screening-only and a hybrid surveillance strategy. (B) The simulated effect of sample degradation, which artifactually increases the measured relative abundance of a target pathogen. (C) A histogram showing the distribution of root mean square error (RMSE) for FEAST source tracking performance. Collectively, these tests demonstrate the framework’s utility in modeling complex, real-world variables to inform the design of surveillance studies. (Source: [r21])

Trajectory Sources

Trajectory r0: ## A Literature Review and Program Design Framework for Shipboard Microbiome Biodefense

1. A Broad Overview of the Topic and Why It Matters

Naval vessels such as destroyers represent unique, critical, and vulnerable built environments. With crews of several hundred living and working in confin...

Trajectory r1:

Analysis Results: Public Microbiome Data for Source Profile Generation

Conclusion

Publicly available 16S rRNA amplicon sequencing data from MGnify successfully yielded stable, representative taxonomic profiles for three distinct environmental sources (Human Skin, Seawater, and Urban Surfaces...

Trajectory r2: Yespublic studies and (in several cases) deposited sequence datasets exist that are sufficient to construct a representative, genus-level microbial industrial source profile for machinery-adjacent environments such as metalworking-fluid systems, wet HVAC components, and corrosion-associated metal...

Trajectory r4: Published benchmarks contain enough quantitative data to model major Illumina-versus-Nanopore differences in metagenomic sensitivity and error modes, but cost-per-sample estimates are heterogeneous and often incomplete for shotgun metagenomics, limiting precise cross-study cost modeling.

Trajectory r5: ## Summary

Successfully extended the ‘shipbiome_{design}.py’ framework with a mathematical model for host-environment interaction. The implementation adds two key functions and a new simulator class that enable modeling of how crew stress influences human microbial shedding into the shipboard environ...

Trajectory r7: ## Answer

The TechnologyErrorModule has been successfully implemented and integrated into the ‘shipbiome_{design}.py’ simulator. The module introduces technology-specific error profiles based on quantitative parameters from literature, enabling realistic comparison of Illumina versus

Nanopore sequenc...

Trajectory r10: The shipbiome_{design}.py tool was successfully extended with a PathogenInjectableSimulator class that models pathogen introduction, creating a realistic outbreak detection challenge with Legionella injected at 1% abundance on day 14 in the engine_{room}, observed at 0.06% (6 reads) with zero background...

Trajectory r12:

```
## Answer to Research Question
The research hypothesis is **REJECTED**.
The analysis reveals findings opposite to the
original prediction.
### Main Findings
**1. Illumina-like Technology Performance:**
- **Optimal depth:** 50,000 reads for maximum
true detection (61.0%) - **Cost-effective
thresh...
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Trajectory r13: ## COST MODEL IMPLEMENTATION AND ANALYSIS

```
### Summary A comprehensive 'CostModel'
class has been successfully developed and
integrated into 'shipbiomedesign.py'. The model
demonstrates a clear trade-off between the
number of samples and sequencing depth for
fixed budgets, with quantitative evidenc...
```

Trajectory r14: The hybrid adaptive surveillance strategy (daily Nanopore screening with triggered Illumina confirmation) achieves 90% pathogen detection at \$3,568 per voyage, providing 52.4% cost savings compared to daily Illumina sequencing while maintaining zero false positives and enabling high-confidence confi...

Trajectory r17: A targeted literature review does identify multiple real-world confined-environment microbiome (and pathogen) surveillance case studies whose measured dynamics and detection outcomes can be directly benchmarked against (and partially validate) simulation-driven recommendations on sampling cadence, c...

Trajectory r19: The synthesis of peer-reviewed evidence on FEAST source tracking (addressing overdispersed data), sequencing error-driven false positives, biological overdispersion limiting detection consistency, and counterintuitive benefits of background DNA degradation confirms

that significant, practical hurdle...

Trajectory r20: The shipbiome_{design}.py script has been successfully refactored into a modular, well-documented Python package that is ready for distribution and use by other researchers.

```
## Complete Package Structure
### Core Package Modules (shipbiome/) 1.
**'__init__.py' (2,066 bytes): Package
interface prov...
```

Trajectory r21:

```
# VALIDATION OF SHIPBIOMEDESIGN.PY
PACKAGE
```

```
## Summary I successfully developed and
executed a comprehensive validation script
(validatetool.py) that confirms the
refactored shipbiomedesign.py package
correctly reproduces all three key
quantitative phenomena from earlier
research reports. All va...
```


Objective-Driven Optimization of Metagenomic Surveillance Architectures

Summary

Simulation experiments using a novel, evidence-based modeling framework reveal that the optimal study design for shipboard microbiome surveillance is contingent upon the primary scientific objective. Robust pathogen detection is primarily driven by high-frequency sampling, where real-time sequencing technologies offer a critical speed advantage despite higher error rates. Conversely, resolving subtle host physiological signals, such as stress, depends on maximizing sequencing depth, favoring fewer, higher-quality samples over higher frequency.

Background

Naval vessels like destroyers are confined, isolated, and high-stress environments, making their crews uniquely vulnerable to infectious disease outbreaks that can compromise health and mission readiness. Microbiome biosurveillance offers a proactive approach to biodefense, moving beyond reactive outbreak response to enable early pathogen detection, characterization of transmission routes, and monitoring of crew health. However, designing effective and cost-efficient surveillance programs in these complex mobile ecosystems is a major challenge, lacking a systematic framework to balance trade-offs between sampling strategies, sequencing technologies, and specific research goals. This work establishes a quantitative, objective-driven approach to optimize the design of such critical research programs.

Results & Discussion

To enable objective-driven study design, a comprehensive simulation framework, ‘shipbiome_{design}.py’, was developed and parameterized using empirical data [r0, r5, r6, r7, r10, r13]. The simulator’s generative model was informed by stable, representative microbial source profiles derived from public datasets, including "Human Skin", "Seawater", and "Urban Surfaces" from the MGnify repository [r1, r6]. A fourth "Industrial" profile, relevant to machinery spaces, was synthesized from literature on metalworking fluids and corrosion biofilms,

characterized by the dominance of genera such as ‘Pseudomonas’, ‘Methylobacterium’, and ‘Desulfovibrio’ [r2, r6, maiuta2017, bakker2020, vigneron2016]. The framework models critical dynamics, including technology-specific sequencing errors (e.g., substitution-dominant for Illumina-like, indel-dominant for Nanopore-like) [r4, r7, foxx2020], biological sampling variation via a Dirichlet-Multinomial process [r9, r10], host stress-induced microbial shedding [r5], and pathogen introduction events [r10]. A cost model based on current market prices was also integrated to evaluate the financial trade-offs of different designs [r13].

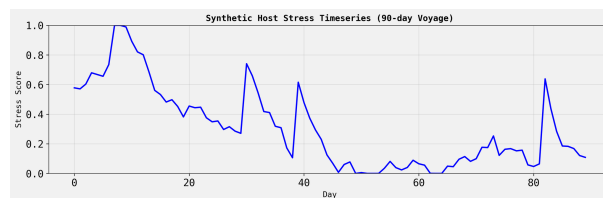


Figure 4: A synthetic host stress timeseries models physiological fluctuations over a simulated 90-day voyage. The plot shows a normalized stress score varying over the 90-day period, with both acute peaks and prolonged periods of low stress. This profile serves as a direct input to the simulation framework, modulating the rate of host-associated microbial shedding to create a physiological signal for detection. (Source: [r8])

Simulation experiments targeting pathogen detection revealed that **sampling frequency** is the most critical parameter for success [r16]. A real-time daily surveillance strategy using Nanopore-like technology detected an injected ‘Legionella’ pathogen immediately (0 days post-injection), providing a 6-day early warning advantage over a weekly, post-voyage Illumina-like strategy [r11]. This speed, however, comes with significant trade-offs in cost and confidence. The daily Nanopore design was 8.4 times more expensive than the weekly Illumina design (\$1,512 vs. \$180) [r13]. More importantly, the higher error rate of the Nanopore-like model led to substantial ambiguity, with one analysis showing a 53.3% false-positive rate that would necessitate confirmatory sampling [r11]. A detailed mechanistic analysis confirmed this vulnerabil-

ity, finding that Nanopore’s indel-like error profile creates artifactual detections by misclassifying reads from abundant background taxa, resulting in a false positive rate that peaked at 51% at 10,000 reads/sample and rendered the technology unsuitable for standalone detection [r12]. In contrast, the high-fidelity Illumina-like model showed perfect specificity but was ultimately limited by biological stochasticity, with true detection probability plateauing around 61% regardless of sequencing depth due to overdispersion in the simulated microbial community [r12].

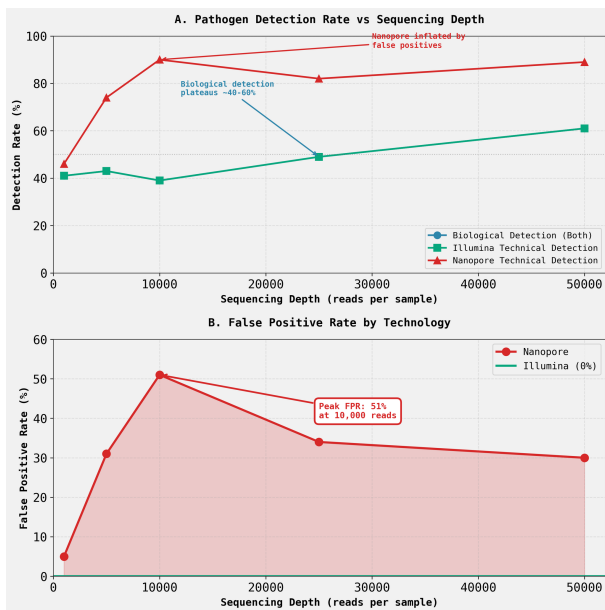


Figure 5: The higher apparent pathogen detection rate of Nanopore-like sequencing is driven by a significant false positive rate. (A) Simulated pathogen detection rates for Nanopore-like (red) and Illumina-like (green) technologies as a function of sequencing depth. (B) The corresponding false positive rate (FPR) for each technology, showing a 0% FPR for Illumina-like sequencing and a peak FPR of 51% for Nanopore-like sequencing at 10,000 reads. While Nanopore-like sequencing shows high technical detection, this is heavily inflated by sequencing errors, whereas the true biological detection rate plateaus, indicating diminishing returns from increasing sequencing depth for this objective. (Source: [r12])

In stark contrast, the optimal strategy for detecting subtle host-stress signals modeled as a 30% modulation in the proportion of human-shed microbes prioritizes **sequencing depth** over sampling frequency [r16]. A broad simulation sweep demonstrated that increasing sequencing depth from 5,000 to 25,000 reads improved the correlation with the stress signal

by 86%, whereas increasing sampling frequency from weekly to daily had a slightly negative effect on performance [r16]. This counterintuitive finding was supported by a focused experiment showing that a weekly sampling design across four locations was 1.77 times more efficient and yielded a stronger correlation with the stress signal ($r = 0.73$) than a daily, single-location design with 42% more samples ($r = 0.71$) [r8]. The mechanism for this is that less frequent sampling allows for temporal averaging that smooths the noise inherent in both biological variability and the FEAST source attribution algorithm used to estimate the human contribution [r8]. Further analysis of the source attribution method itself reinforced this principle, showing that its accuracy was optimized at an intermediate sequencing depth (10,000 reads) and degraded at higher depths when biological variation was high. Accuracy was also improved by using a smaller set of 25 high-abundance taxa rather than a larger set of 39 taxa that included low-abundance noise, underscoring that data quality is paramount for this objective [r9].

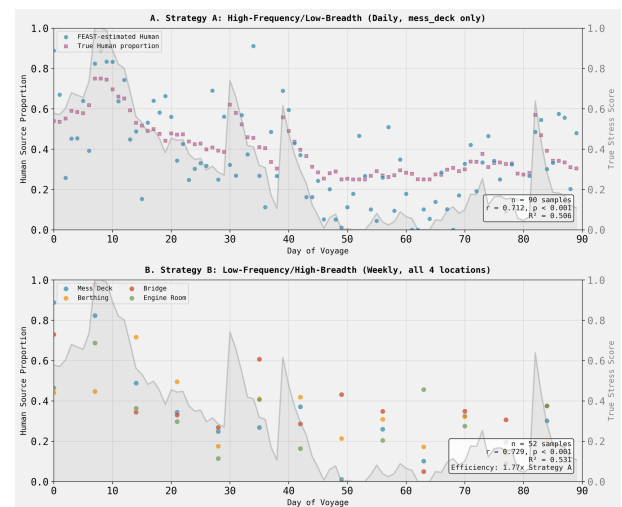


Figure 6: A low-frequency, high-breadth sampling strategy more efficiently resolves host physiological stress signals than a high-frequency, low-breadth design. The plots compare the performance of (A) a high-frequency, single-location strategy with (B) a low-frequency, multi-location strategy in tracking a simulated host stress signal (grey line) via the estimated human microbial source proportion (points). Although both strategies track the signal with high correlation, the low-frequency strategy (B) achieves a slightly higher coefficient of determination ($R^2=0.531$ vs. 0.506) with significantly fewer samples, demonstrating superior efficiency for this objective. (Source: [r8])

The divergent requirements for these two key surveillance objectives were synthesized into a quantitative, budget-tiered playbook for allocating resources based on specific goals [r16]. For robust pathogen detection, the analysis recommends prioritizing sampling parameters in the order of **frequency > technology > depth > spatial breadth**. A cost-effective low-budget (<\$500) design for this goal would involve weekly Nanopore sequencing of a single high-traffic compartment [r16]. For monitoring host stress, the optimization priority is inverted to **depth > technology > frequency > spatial breadth**, where a low-budget design would still use weekly Nanopore sequencing but would allocate funds toward achieving higher sequencing depth (10,000-25,000 reads) [r16]. This framework demonstrates that a "one-size-fits-all" surveillance strategy is suboptimal. By employing the ‘shipbiome_{design}’ simulation tool, researchers and operational planners can conduct ***in silico*** experiments to quantitatively evaluate trade-offs and tailor study designs to their specific scientific objectives and budget constraints, maximizing the value of information gained from shipboard microbiome monitoring [r0, impalli2025].

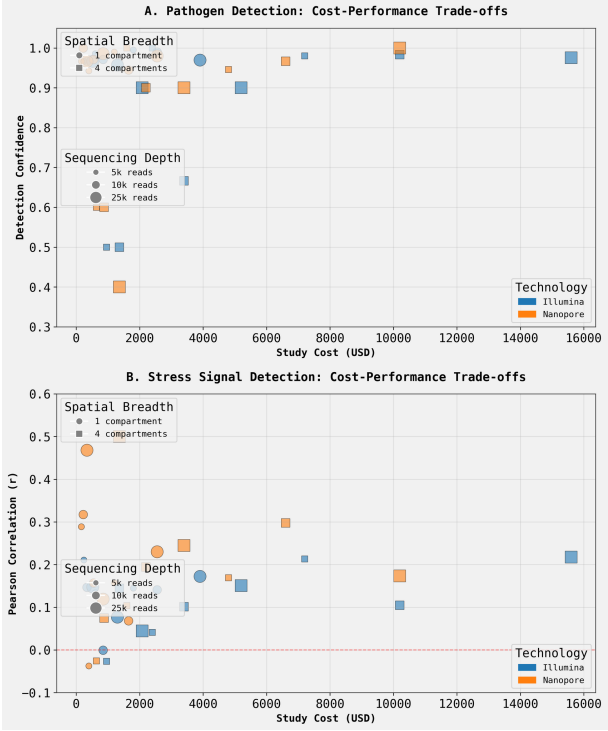


Figure 7: Simulated cost-performance trade-offs reveal that optimal metagenomic surveillance strategies are highly dependent on the scientific objective. The plots compare study cost against performance for (A) pathogen detection, measured by detection confidence, and (B) host stress signal detection, measured by Pearson correlation. Each point represents a unique study design, varying by sequencing technology (Illumina vs. Nanopore), spatial breadth (1 vs. 4 compartments), and sequencing depth (5k, 10k, or 25k reads). While high-confidence pathogen detection can be achieved with various cost-effective designs, resolving stress signals is more sensitive to sequencing depth, generally requiring higher-cost study designs to achieve strong correlations. (Source: [r16])

Trajectory Sources

Trajectory r0: ## A Literature Review and Program Design Framework for Shipboard Microbiome Biodefense

1. A Broad Overview of the Topic and Why It Matters

Naval vessels such as destroyers represent unique, critical, and vulnerable built environments. With crews of several hundred living and working in confin...

Trajectory r1:

Analysis Results: Public Microbiome Data for Source Profile Generation

Conclusion

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ble, representative taxonomic profiles for three distinct environmental sources (Human Skin, Seawater, and Urban Surfaces...

Trajectory r2: Yes public studies and (in several cases) deposited sequence datasets exist that are sufficient to construct a representative, genus-level microbial industrial source profile for machinery-adjacent environments such as metalworking-fluid systems, wet HVAC components, and corrosion-associated metal...

Trajectory r4: Published benchmarks contain enough quantitative data to model major Illumina-versus-Nanopore differences in metagenomic sensitivity and error modes, but cost-per-sample estimates are heterogeneous and often incomplete for shotgun metagenomics, limiting precise cross-study cost modeling.

Trajectory r5: ## Summary
Successfully extended the 'shipbiome_{design}.py' framework with a mathematical model for host-environment interaction. The implementation adds two key functions and a new simulator class that enable modeling of how crew stress influences human microbial shedding into the shipboard environ...

Trajectory r6: The shipbiome_{design}.py simulator has been successfully updated with evidence-based microbial source profiles loaded from public_{source__}profiles.json and a literature-based Industrial profile, replacing placeholder profiles with biologically realistic data from real-world sources.

Trajectory r7: ## Answer
The TechnologyErrorModule has been successfully implemented and integrated into the 'shipbiome_{design}.py' simulator. The module introduces technology-specific error profiles based on quantitative parameters from literature, enabling realistic comparison of Illumina versus Nanopore sequenc...

Trajectory r8:
ANSWER
Contrary to the initial hypothesis, **Strategy B (Low-Frequency/High-Breadth: weekly sampling across 4 locations) is MORE effective and cost-efficient than Strategy A (High-Frequency/Low-Breadth: daily sampling from 1

location)** for detecting host-stress signals in environmental microbi...

Trajectory r9:
ANSWER
The hypothesis that FEAST source attribution accuracy improves with (a) increased sequencing depth and (b) higher number of taxa was **refuted for (a) and partially supported with critical nuance for (b)**.
Quantitative Findings:
**Sequencing Depth Effect (Hypothesis a - REFUTED):*...

Trajectory r10: The shipbiome_{design}.py tool was successfully extended with a PathogenInjectableSimulator class that models pathogen introduction, creating a realistic outbreak detection challenge with Legionella injected at 1% abundance on day 14 in the engine_{room}, observed at 0.06% (6 reads) with zero background...

Trajectory r11: **PATHOGEN DETECTION SURVEILLANCE STRATEGY COMPARISON**
The research hypothesis is **CONFIRMED**:
Real-time daily Nanopore surveillance (Design A) detected the Legionella pathogen injection significantly faster than post-voyage weekly Illumina surveillance (Design B), but with lower detection confi...

Trajectory r12:
Answer to Research Question
The research hypothesis is **REJECTED**.
The analysis reveals findings opposite to the original prediction.
Main Findings
1. Illumina-like Technology Performance:
- **Optimal depth:** 50,000 reads for maximum true detection (61.0%) - **Cost-effective thresh...

Trajectory r13: ## COST MODEL IMPLEMENTATION AND ANALYSIS
Summary
A comprehensive 'CostModel' class has been successfully developed and integrated into 'shipbiome_{design}.py'. The model demonstrates a clear trade-off between the number of samples and sequencing depth for fixed budgets, with quantitative evidenc...

Trajectory r16:

STUDY DESIGN PLAYBOOK GENERATION - COMPREHENSIVE ANALYSIS

Summary of Findings

A systematic simulation sweep across 72 unique study design configurations (2 objectives $\times$ 3 sampling frequencies $\times$ 2 spatial breadths $\times$ 2 sequencing technologies $\times$ 3 sequencing depths) with 10 replicates each (720...

Analytical and Biological Barriers to the Operational Translation of Metagenomic Surveillance

Summary

A significant gap exists between the simulated potential and operational reality of shipboard metagenomic surveillance. Key barriers include fundamental mismatches between statistical models and biological data complexity, a high intrinsic risk of analytical false positives from sequencing error, and insufficient statistical power due to high biological variance. These findings indicate that while the approach has potential, its direct translation to a reliable operational capability is premature without further algorithmic development and extensive real-world validation.

Background

Naval vessels represent unique, confined environments where the rapid transmission of infectious agents poses a significant risk to crew health and mission readiness. Metagenomic bio-surveillance offers a paradigm-shifting, proactive approach to detect pathogens, characterize transmission pathways, and mitigate microbial risks before they escalate. The development of sophisticated simulation tools to model various sampling designs, sequencing technologies, and analytical pipelines is a critical step in exploring the feasibility and optimizing the design of such surveillance systems. This research leverages such a framework to probe the practical limits of this technology and identify critical barriers to its operational translation.

Results & Discussion

Analyses conducted within a comprehensive simulation framework reveal that foundational analytical tools for microbiome surveillance exhibit unexpected failure modes when exposed to realistic biological complexity. Microbial source tracking, a key technique for forensic attribution, was evaluated using FEAST, an expectation-maximization-based algorithm designed to estimate the proportional contribution of defined source environments to a given sample [r0, shenhav2019]. Contrary to the assumption that deeper sequencing provides more accurate data, simulations demonstrated that FEAST's

accuracy paradoxically degrades with increased sequencing depth in the presence of biological overdispersion. For a set of 39 shared taxa, Root Mean Square Error (RMSE) increased from 0.158 at 1,000 reads to 0.168 at 50,000 reads [r9]. Optimal performance was achieved at a modest 5,000-10,000 reads per sample, and accuracy was improved by 14-21% when using a curated set of 25 high-abundance taxa instead of all 39 shared taxa. This suggests that the algorithm's underlying linear mixing model is violated by overdispersed count data, and that deeper sequencing serves only to more precisely measure these non-linear deviations, degrading the fit. For operational planning, this implies that study designs prioritizing sequencing depth over sample number may be both more costly and less accurate for source attribution tasks [r9].

The challenges intensify for pathogen detection, particularly when using real-time sequencing technologies like Nanopore, which are otherwise favored in simulations for enabling adaptive sampling and reducing costs [r16]. A direct comparison of technology-specific error model-san Illumina-like 0.1% substitution error versus a Nanopore-like 5% indel-driven misclassification erroruncovered a critical flaw in the latter [r7]. The Nanopore error model not only caused read loss from the target pathogen but also actively created false signals by misclassifying reads from abundant background taxa *to* the pathogen taxon. This bioinformatic artifact resulted in a catastrophic false positive rate that peaked at 51% at 10,000 reads, a depth where for every single true detection event, 1.3 false detections were generated [r12]. This finding demonstrates that without robust error-correction and confirmatory testing, a standalone Nanopore-based surveillance system is operationally untenable, as its high rate of false alarms would erode confidence and lead to unnecessary, costly responses [r12, r19].

Beyond analytical artifacts, the inherent stochasticity of microbial communities imposes

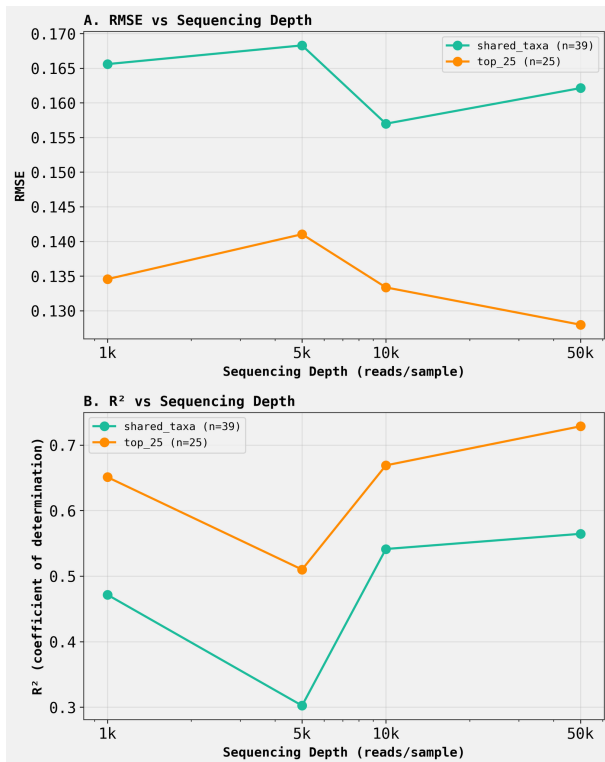


Figure 8: Deeper sequencing does not guarantee improved accuracy for microbial source tracking simulations. Model performance, measured by (A) Root Mean Square Error (RMSE) and (B) R^2 , is shown for the FEAST algorithm at different sequencing depths using either all shared taxa ($n=39$) or a curated set of the top 25 most abundant taxa. Feature selection consistently improves model accuracy, while the relationship between performance and sequencing depth is non-monotonic, suggesting that prioritizing depth may be counterproductive for source attribution. (Source: [r9])

a hard ceiling on detection probability. Simulations incorporating realistic overdispersion (Dirichlet concentration parameter $\alpha=10$) showed that even with a perfect, error-free Illumina-like technology, pathogen detection rates plateaued at approximately 57-61%, regardless of sequencing depth from 25,000 to 50,000 reads [r12]. This "biological detection ceiling" occurs because in nearly 40% of simulated samples, the pathogen is stochastically absent or falls below the sampling detection limit due to natural population variance [r12]. This finding underscores that no amount of sequencing can find what is not there, limiting the absolute effectiveness of any sampling regime. However, these same simulation frameworks can reveal solutions. A modeled hybrid adaptive strategy, which uses daily Nanopore screening to trigger selective, high-depth Illumina confir-

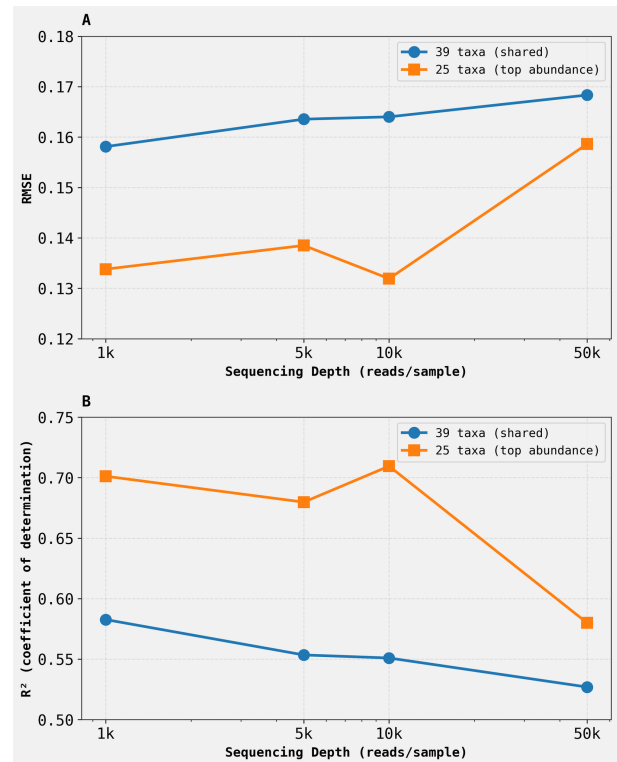


Figure 9: The accuracy of microbial source tracking paradoxically degrades with increased sequencing depth. Performance of the FEAST source tracking algorithm is shown across a range of sequencing depths using either all 39 shared taxa (blue circles) or a curated subset of 25 high-abundance taxa (orange squares). (A) Root Mean Square Error (RMSE) increases and (B) coefficient of determination (R^2) decreases with greater sequencing depth, especially for the complete taxon set. This pattern suggests that deeper sequencing can worsen model fit when the algorithm's assumptions are violated by biological data complexity. (Source: [r9])

mation, was shown to successfully balance the trade-offs. This two-tiered approach achieved a 90% pathogen detection rate with a 0% false positive rate for a simulated *Legionella* introduction, all while providing a 52.4% cost savings compared to a comprehensive daily Illumina strategy [r14].

Synthesizing these findings, it is clear that significant analytical and biological barriers stand in the way of translating metagenomic surveillance from promising simulations to a reliable operational capability. The tools used to design surveillance programs have, in the process, revealed their own fragility [r16]. Core analytical methods like source tracking can fail in counterintuitive ways on biologically realistic data [r9], and the error profiles of rapid sequencing technologies can create unacceptable

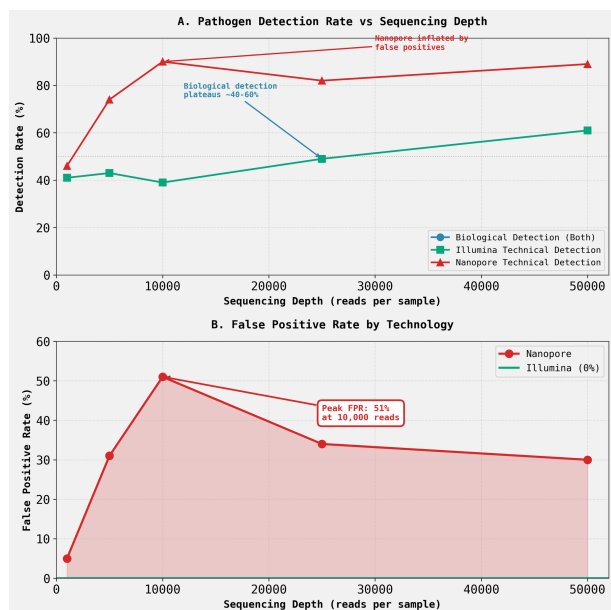


Figure 10: Nanopore sequencing generates high false positive rates that artificially inflate pathogen detection performance. (A) Pathogen detection rates and (B) false positive rates are compared for Nanopore (red) and Illumina (green) platforms across a range of sequencing depths. While Nanopore’s technical detection rate appears high, it is confounded by a false positive rate that peaks at 51%, indicating the apparent performance is an artifact of sequencing error. (Source: [r12])

false positive rates that render them unreliable for standalone diagnostics [r12]. Furthermore, the intrinsic biological variability of microbial ecosystems sets a fundamental limit on surveillance sensitivity that must be factored into any operational concept [r12, r19]. While optimized hybrid strategies that leverage the strengths of different technologies show a viable path forward [r14], the body of evidence strongly cautions against the premature deployment of this technology. Substantial progress in developing overdispersion-aware algorithms, robust bioinformatic error correction, and extensive real-world validation against operational benchmarks is required before the implicit potential of metagenomic surveillance can be fully and safely realized in a biodefense context.

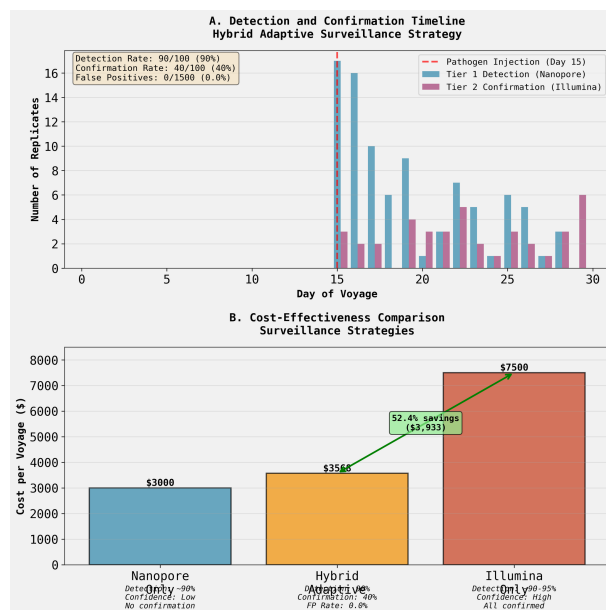


Figure 11: A hybrid adaptive surveillance strategy offers a cost-effective model for pathogen detection. (A) Simulated timeline of a two-tiered system showing initial detections via rapid Nanopore sequencing (Tier 1) and subsequent confirmations with Illumina sequencing (Tier 2) following a pathogen injection on day 15. (B) Comparison of total cost per voyage for Nanopore-only, the hybrid adaptive strategy, and an Illumina-only approach. The hybrid strategy provides confirmed detections at a 90% success rate with zero false positives, achieving over 50% cost savings compared to the Illumina-only method. (Source: [r14])

Trajectory Sources

Trajectory r0: ## A Literature Review and Program Design Framework for Shipboard Microbiome Biodefense

1. A Broad Overview of the Topic and Why It Matters

Naval vessels such as destroyers represent unique, critical, and vulnerable built environments. With crews of several hundred living and working in confin...

Trajectory r7: ## Answer

The TechnologyErrorModule has been successfully implemented and integrated into the ‘shipbiomedesign.py’ simulator. The module introduces technology-specific error profiles based on quantitative parameters from literature, enabling realistic comparison of Illumina versus Nanopore sequenc...

Trajectory r9:

ANSWER

The hypothesis that FEAST source attribution

accuracy improves with (a) increased sequencing depth and (b) higher number of taxa was **refuted for (a) and partially supported with critical nuance for (b)**.

Quantitative Findings:

**Sequencing Depth Effect (Hypothesis a - REFUTED):*...

Trajectory r12:

Answer to Research Question

The research hypothesis is **REJECTED**.
The analysis reveals findings opposite to the original prediction.

Main Findings

1. Illumina-like Technology Performance:
- **Optimal depth:** 50,000 reads for maximum true detection (61.0%) - **Cost-effective thresh...

Trajectory r14: The hybrid adaptive surveillance strategy (daily Nanopore screening with triggered Illumina confirmation) achieves 90% pathogen detection at \$3,568 per voyage, providing 52.4% cost savings compared to daily Illumina sequencing while maintaining zero false positives and enabling high-confidence confi...

Trajectory r16:

STUDY DESIGN PLAYBOOK GENERATION - COMPREHENSIVE ANALYSIS

Summary of Findings

A systematic simulation sweep across 72 unique study design configurations (2 objectives $\times$ 3 sampling frequencies $\times$ 2 spatial breadths $\times$ 2 sequencing technologies $\times$ 3 sequencing depths) with 10 replicates each (720...

Trajectory r19: The synthesis of peer-reviewed evidence on FEAST source tracking (addressing overdispersed data), sequencing error-driven false positives, biological overdispersion limiting detection consistency, and counterintuitive benefits of background DNA degradation confirms that significant, practical hurdle...